



B.Tech. Degree VII Semester Special Supplementary Examination March 2023

ME 19-205-0706 AUTOMOBILE ENGINEERING
(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

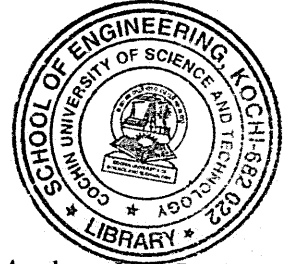
Course Outcome

On successful completion of the course, the students will be able to:

- CO1: Compare the working of various types of automobiles.
 CO2: Learn the functions and working of different engine components.
 CO3: Appreciate the use of alternate fuels.
 CO4: Understand the latest technology used in the automobile field.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate,
 L6 – Create

PO – Programme Outcome



PART A (Answer *ALL* questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a) How do you classify automobile engines?		3	L1	1	1
(b) How is variable valve timing done?		3	L2	1	1
(c) How MPFI system works effectively related to conventional method of fuel injection?		3	L2	2	1
(d) Compare CNG with conventional fuels.		3	L2	2	1
(e) How is camber helpful in steering?		3	L2	3	1
(f) Explain antilock braking system.		3	L1	3	1
(g) What are fuel cells?		3	L2	4	1
(h) How is an electric vehicle transmission structured?		3	L1	4	1

PART B

(4 × 12 = 48)

II.	(a)	What is piston slap and how do you control piston slap?	4	L2	1	1
	(b)	Explain the valve layout diagram of a four stroke cycle SI engine with a neat diagram.	8	L1	1	1
OR						
III.	(a)	Explain various engine parts with neat diagrams.	8	L2	1	1
	(b)	What are the factors to be considered for engine balancing?	4	L1	1	1
IV.	(a)	Compare battery ignition system with magneto ignition system.	6	L1	2	1
	(b)	Explain pressure cooling system in automobiles with the properties required for a good coolant.	6	L2	2	1
OR						
V.	(a)	What are the objectives of a lubrication system?	4	L1	2	1
	(b)	Compare mist lubrication system with wet sump lubrication system with neat sketches.	8	L2	2	1

(P.T.O.)

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		Marks	BL	CO	PO
VI.	(a) Explain a synchromesh gear system with neat figures.	8	L1	3	2
	(b) Explain the different factors for wheel alignment.	4	L2	3	1
OR					
VII.	(a) With neat sketches explain the working of a pneumatic brake system.	8	L2	3	1
	(b) Give an overview of an automobile steering system.	4	L1	3	1
VIII.	(a) How do you classify electrified power trains?	6	L1	4	2
	(b) Comment on the various storage devices used in an electric vehicle.	6	L2	4	1
OR					
IX.	(a) Compare electric vehicles with conventional IC engine driven vehicles.	4	L1	4	1
	(b) Compare a hybrid system with an electric motor system.	8	L2	4	1

Blooms's Taxonomy Levels

L1 – 44.1%, L2 – 55.83%.
